

A SYNOPSIS

ON

**RESEARCHOS: AI-BASED RESEARCH PAPER ANALYSIS AND
COMPARISON SYSTEM**

Submitted in the Partial Fulfilment of the Requirement for

BACHELOR OF TECHNOLOGY

Department of Artificial Intelligence

Faraz Ahmed Siddiqui [54]

Branch : CSE-AIML

UNDER THE GUIDANCE OF

Dr. Amol Chaudhary

DEPARTMENT OF ARTIFICIAL INTELLIGENCE

G H RAISONI COLLEGE OF ENGINEERING, NAGPUR

ACADEMIC YEAR 2026-27 (WINTER)

ABSTRACT

Research students and engineering students often need to examine a large number of research papers before starting a project, thesis, or literature review. The required information is distributed across sections such as the abstract, methodology, experiments, results, and conclusion, making manual extraction and comparison time-consuming and inconsistent. ResearchOS is proposed as an AI-based research paper analysis and comparison system that accepts multiple research-paper PDFs, extracts their textual content, identifies important research information using an LLM-based NLP pipeline, generates structured summaries, and presents a field-by-field comparison of the selected papers. The system focuses on eight standard parameters: Research Problem, Method Used, Dataset, Algorithm, Results, Advantage, Limitation, and Future Scope. The proposed implementation uses React.js for the dashboard, FastAPI and Python for backend services, PyMuPDF/pdfplumber for PDF processing, an LLM through the Groq API for information extraction, and PostgreSQL for structured storage. The system is intended to provide a simple, low-cost, student-oriented workflow from PDF upload to research comparison. It can reduce repetitive literature-review work and improve the consistency of research notes while keeping the final interpretation under the user's control.

KEYWORDS: Natural Language Processing, Large Language Models, PDF Parsing, Text Summarization, Literature Review Automation, Research Paper Comparison

1. INTRODUCTION

Research papers are an essential source of knowledge for students, researchers, and engineers. Before developing a new project, researchers normally study previous work to understand the problem already addressed, methodologies used, datasets, algorithms, experimental results, limitations, and possible future directions. A literature review therefore requires not only finding relevant papers but also extracting comparable information from each paper.

The manual process becomes difficult when several papers have to be reviewed. Important information may appear in different sections and formats, while papers from different journals and conferences may use different terminology and layouts. As the number of papers increases, maintaining consistent notes and constructing a comparison table becomes repetitive and error-prone.

ResearchOS is proposed to automate this workflow. The system allows a user to upload one to five research-paper PDFs, extracts their text, cleans and organizes the content, and uses a Large Language Model (LLM) to identify eight predefined research parameters. It then generates a structured summary for each paper and a side-by-side comparison across multiple papers. The overall workflow is designed as a simple web dashboard so that students can use the system without manually preparing comparison tables.

The project combines document processing, Natural Language Processing, information extraction, text summarization, database management, and web application development into one end-to-end system. The intended outcome is not to replace human research judgment, but to reduce repetitive information-gathering work and provide a consistent starting point for literature analysis.

2. MOTIVATION

The motivation for ResearchOS comes from the practical difficulty faced by students while preparing project reports, research proposals, seminar reports, and literature reviews. A single review may involve dozens of papers, and manually recording the same categories for every paper requires considerable time.

- Key information is scattered across the abstract, methodology, experiments, results, and conclusion.
- Different papers use different structures and terminology, making manual extraction inconsistent.
- Comparing methods, datasets, algorithms, results, advantages, and limitations across papers requires repeated switching between documents.
- Manual notes can contain omissions or inconsistent descriptions.
- Existing research tools are often focused on discovery, citation management, individual-paper reading, or broad literature-review workflows rather than the exact eight-field comparison workflow proposed here.

ResearchOS is therefore motivated by the need for a focused, student-friendly tool that converts a set of uploaded PDFs into structured research notes and a comparison table. The project targets automation of repetitive tasks while retaining a human researcher as the final decision-maker.

3. LITERATURE SURVEY

The literature survey for ResearchOS covers scientific-document summarization, scientific-text representation, document-level similarity, and AI-assisted literature-review workflows. Earlier research shows that scientific articles have specialized structure and contain important information in sections, tables, figures, and citations. More recent work demonstrates the usefulness of transformer-based and LLM-based methods for summarization and scientific-document analysis.

S. No.	Study / Year	Objective / Method	Key Observation / Limitation
1	Cohan et al., 2018 NAACL-HLT	Discourse-aware abstractive summarization of long documents such as research papers using a hierarchical encoder and attention-based decoder.	Shows that scientific papers require models capable of handling long-document structure; computationally more complex than simple extractive methods.
2	Beltagy, Lo & Cohan, 2019 EMNLP-IJCNLP	SciBERT: pretrained language model designed for scientific text.	Scientific-domain pretraining improves representation for scientific NLP tasks, but using a specialized pretrained model adds model and deployment complexity.
3	Cohan et al., 2020 ACL	SPECTER generates document-level scientific-paper embeddings using citation-informed learning.	Useful for document-level similarity and recommendation, but does not by itself produce the structured eight-field comparison required by ResearchOS.
4	Automatic Summarization of Scientific Articles: A Survey, 2020/2022	Reviews automatic scientific-article summarization methods and highlights the special structure of scientific literature.	Confirms that scientific summarization differs from generic summarization and must account for paper-specific structure.
5	Comparative Analysis of LLMs for Scientific Article Summarization, 2025	Compares GPT-3.5, LLaMA 3, Mixtral 8x7B and Gemma-2 for scientific article summarization using ROUGE, BERT and BLEU alongside qualitative assessment.	Demonstrates the growing role of LLMs in scientific summarization and the need for both quantitative and qualitative evaluation.

From the survey, three observations are important for the proposed system. First, scientific documents require structure-aware processing. Second, transformer/LLM-based approaches can generate useful summaries and extract semantic information. Third, document-level similarity alone is not sufficient when a researcher wants to compare concrete fields such as methodology, dataset, algorithm, results, advantages, limitations, and future scope. ResearchOS combines these ideas into a single structured workflow.

4. EXISTING SYSTEMS

Several existing platforms support parts of the research workflow. They are useful references for the design of ResearchOS, but their primary workflows differ from the focused upload-and-compare objective of this project.

ResearchOS is not intended to compete with large research platforms in paper discovery or systematic-review scale. Its contribution is a focused end-to-end workflow for engineering students: upload a small batch of PDFs, extract the same research parameters from every paper, generate concise summaries, and compare the results in a consistent format.

System	Main Capability	Limitation with respect to ResearchOS
Semantic Scholar	AI-powered scientific literature search, paper discovery, recommendations, library management and AI-assisted paper understanding.	Primarily a literature discovery and reading ecosystem; the proposed project focuses on extracting the same predefined fields from a user-uploaded batch and comparing them in one table.
Elicit	AI-assisted scientific research, literature search, reports, systematic-review workflows and data extraction.	Broader and more advanced research workflow; ResearchOS is intentionally narrower and designed as a small, student-built local/web prototype with a fixed eight-field schema.
SciSpace	Literature search, literature-review support, Chat with PDF, extraction and other research tools.	Provides a broad commercial research environment; ResearchOS focuses specifically on a transparent eight-parameter comparison pipeline for a small number of uploaded PDFs.
ResearchRabbit	Research discovery, citation connections, paper maps and literature exploration.	Strong on discovery and visualization of research connections; it is not designed around the fixed field-by-field extraction and comparison workflow proposed here.

5. OBJECTIVES

The main objective of ResearchOS is to develop an AI-assisted system that simplifies the analysis and comparison of research papers. The specific objectives are:

- To develop a web-based interface for uploading one to five research-paper PDFs.
- To reliably extract textual content from uploaded PDF documents using PyMuPDF and/or pdfplumber.
- To clean, organize, section, and prepare extracted text for downstream NLP processing.
- To use an LLM-based information-extraction pipeline to identify eight standard research parameters from each paper.
- To generate a structured summary for every uploaded paper.
- To automatically align the same parameters across multiple papers and produce a side-by-side comparison table.
- To store paper metadata and extracted information in a PostgreSQL database.
- To evaluate the system using extraction accuracy, precision/recall/F1 where applicable, summary quality, and processing time.
- To provide a simple and low-cost tool suitable for engineering students and academic project work.

6. SCOPE OF THIS PROJECT

The initial scope is deliberately limited so that the system can be implemented, tested, and demonstrated as a third-semester engineering minor project.

In Scope

- Upload and process 1–5 PDF research papers per session.
- Extract research problem, method used, dataset, algorithm, results, advantage, limitation, and future scope.
- Generate a structured summary for each paper.
- Generate a comparison table for 2–5 papers.
- Provide a basic dashboard for upload, processing status, summaries, and comparison.
- Store paper metadata and extracted information in PostgreSQL.
- Optional future features include exportable PDF/Excel reports and similarity scoring.

Out of Scope for Phase 1

- User accounts and authentication.
- Mobile application development.
- Multi-language research-paper support.
- Payment or subscription functionality.
- Real-time collaborative editing.
- Fully autonomous systematic literature review or replacement of human research judgment.

7. IMPLEMENTATION OF PROPOSED SYSTEM

The proposed system follows a modular pipeline in which document processing, AI analysis, database storage, and presentation are separated. This makes the system easier to test and extend.

Upload PDFs → Validate files → Extract text → Clean and section text → Chunk long content if required → Send structured prompts to LLM → Parse the response into the eight-field schema → Validate/store extracted data →

Generate paper summary → Compare selected papers → Display results on dashboard.

7.1 Architecture of Proposed System

Stage	Function
1. User	Uploads 1–5 research-paper PDFs
2. React Dashboard	Sends files and displays processing status, summaries and comparisons
3. FastAPI Backend	Receives uploads, manages processing requests and exposes REST APIs
4. PDF Processing	PyMuPDF / pdfplumber extracts text and document metadata
5. Preprocessing	Cleans text, removes unnecessary artifacts and organizes sections/chunks
6. LLM Analysis	Groq API + LLM extracts the eight predefined research parameters
7. PostgreSQL	Stores paper metadata and structured extracted data
8. Output Engine	Generates per-paper summary and field-by-field comparison

7.2 Method / Algorithms / Tools Used

- PDF Processing: PyMuPDF and pdfplumber for text extraction and document handling.
- Preprocessing: text cleaning, section detection, chunking and metadata preparation.
- NLP / AI: LLM-based prompt-guided information extraction and summarization through the Groq API.
- Structured Information Extraction: fixed schema containing the eight project parameters.
- Comparison: field-by-field alignment of extracted values across papers.
- Backend: Python and FastAPI.
- Frontend: React.js, HTML, CSS and JavaScript.
- Database: PostgreSQL using Supabase/Neon free-tier infrastructure.
- Visualization: Plotly/D3.js where required for charts or similarity views.
- Evaluation: extraction accuracy, precision, recall, F1, ROUGE-L and processing time.

7.3 Proposed Processing Flow

Upload PDFs → Validate files → Extract text → Clean and section text → Chunk long content if required → Send structured prompts to LLM → Parse the response into the eight-field schema → Validate/store extracted data → Generate paper summary → Compare selected papers → Display results on dashboard.

7.4 Database Structure

The planned database contains a papers table for file metadata and processing status, and an extracted_data table linked to each paper. The extracted_data table stores research_problem, method_used, dataset, algorithm, results, advantage, limitation, and future_scope. Keeping these field names consistent across the database, prompts, APIs, and interface reduces integration errors.

7.5 Planned API Endpoints

- POST /upload – upload PDF file(s).
- GET /papers – list uploaded papers.
- GET /papers/{id} – retrieve paper details.
- POST /papers/{id}/analyze – run AI extraction.
- GET /papers/{id}/summary – retrieve structured summary.

- POST /compare – compare selected papers.
- GET /papers/{id}/status – check processing status.

8. CONCLUSION

ResearchOS proposes an end-to-end AI-assisted workflow for analyzing and comparing research papers. Instead of requiring students to manually read every paper and prepare separate notes before building a comparison table, the system will automate the repetitive stages of PDF text extraction, information extraction, structured summarization, and cross-paper comparison. The proposed architecture combines PDF-processing libraries, an LLM-based NLP service, a relational database, and a web dashboard.

The initial implementation is intentionally focused on five or fewer papers and eight clearly defined research parameters. This bounded scope makes the system suitable for an engineering minor project while still demonstrating practical applications of Natural Language Processing, Large Language Models, document intelligence, backend development, database management, and frontend integration. The expected result is a consistent and time-saving research assistant that supports students during literature review preparation without replacing human evaluation of the original papers.

Future extensions may include larger batch processing, citation-aware similarity analysis, OCR for scanned papers, export to PDF/Excel, multilingual support, searchable paper libraries, and evidence-linked extraction in which each generated field is accompanied by the source passage from the original paper.

9. REFERENCES

1. A. Cohan, F. Dernoncourt, D. S. Kim, T. Bui, S. Kim, W. Chang, and N. Goharian, “A Discourse-Aware Attention Model for Abstractive Summarization of Long Documents,” *Proceedings of NAACL-HLT*, 2018, pp. 615–621.
2. I. Beltagy, K. Lo, and A. Cohan, “SciBERT: A Pretrained Language Model for Scientific Text,” *Proceedings of EMNLP-IJCNLP*, 2019, pp. 3615–3620.
3. A. Cohan, S. Feldman, I. Beltagy, D. Downey, and D. Weld, “SPECTER: Document-level Representation Learning using Citation-informed Transformers,” *Proceedings of ACL*, 2020, pp. 2270–2282.
4. “Automatic summarization of scientific articles: A survey,” *Journal of King Saud University Computer and Information Sciences*, published online 2020; volume 34, pp. 1011–1028, 2022.
5. “Comparative Analysis of Large Language Models for the Application of Scientific Article Summarization,” *Procedia Computer Science*, vol. 259, pp. 532–542, 2025, DOI: 10.1016/j.procs.2025.04.002.
6. PyMuPDF Documentation, “Text Extraction and PDF Processing,” Artifex Software, 2026.
7. Semantic Scholar, “AI-Powered Research Tool for Scientific Literature,” Allen Institute for AI, 2026.
8. Elicit, “AI for Scientific Research,” Elicit, 2026.
9. SciSpace, “AI for Research and Literature Review,” SciSpace, 2026.
10. ResearchRabbit, “AI Tool for Smarter, Faster Literature Reviews,” ResearchRabbit, 2026.